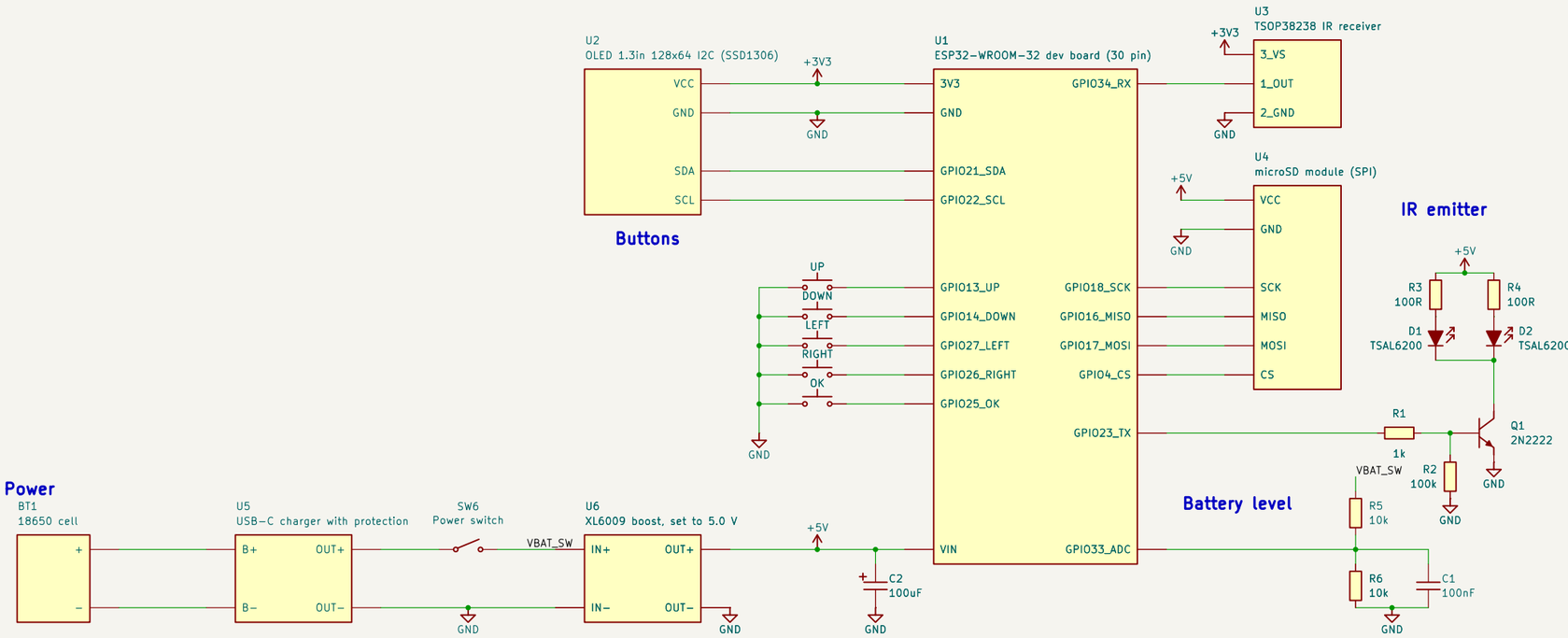


Notes

GPIO numbers are those of include/board_pins.h. GPIO35 (second receiver) and GPIO19 (second emitter) are reserved, not connected.

Buttons are active low: internal pull-ups, no external resistors. The five switches share one ground line.

Power: the switch sits before the XL6009, so the divider stops draining the cell when the unit is off. Charge only with the switch off (no power path).

Emitter: about 34 mA per LED was calculated from the resistor values, not measured. The TSAL6200 is rated 100 mA continuous.

Pin order of the OLED, the microSD module and the TSOP varies between parts: read the silkscreen. TSOP38238 seen from the dome side, legs down: 1 OUT, 2 GND, 3 VS.

The battery divider needs the firmware option IRIS_BATTERY_RATIO_PERMILLE=2000 (set in the esp32dev_ssd1306 environment).

Generated; the modules are drawn as blocks with the pins of the author's parts
Drawn from docs/breadboard-wiring.md and include/board_pins.h
github.com/nruizneiman/IRis

Sheet: /
File: IRis.kicad_sch

Title: IRis – connection diagram

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